

Ambient-Powered Computing

Why the End of the Battery Is a Systems Problem, Not a Power Problem

Abstract

Wireless power research has spent decades optimizing one variable: how much energy can be transmitted, and how precisely it can be delivered. That framing is the wrong problem.

Ambient-powered computing — devices that harvest continuous, diffuse electromagnetic energy without alignment, docking, or beam-tracking — becomes inevitable through the convergence of four independent curves: device power consumption falling by orders of magnitude, rectenna harvesting efficiency rising, compute decentralizing away from the device into data-center and AI infrastructure that big tech is already financing at a scale no single application justifies, and launch costs collapsing under reusable rocket economics until orbital infrastructure becomes financially trivial.

The right question is not *how much power can be delivered*. It is *when does harvested power exceed what a device actually needs — and when does it become economical to beam that power down from orbit at planetary scale*? Three of these four curves are already moving, for reasons that have nothing to do with wireless power. This paper argues that ambient orbital power is not a distant speculative technology. It is the default outcome of four capital-driven trends that are already underway, and connecting them requires no leap of faith — only the willingness to look at four industries simultaneously instead of one at a time

1. Introduction

Modern computing has advanced along nearly every axis imaginable. One component has not kept pace: **the battery**.

Battery energy density improves slowly, on a timeline disconnected from the rest of computing. It remains the bottleneck constraining device size, lifespan, placement, and scale — and it will remain the bottleneck for as long as devices are architected around the assumption that power must be stored locally rather than harvested continuously.

This paper rejects that assumption.

Ambient-powered computing assumes devices harvest usable energy from the electromagnetic environment that already surrounds them, continuously, without alignment or tracking. This is a different bet than the one most wireless power research makes. The dominant approach asks how to deliver more power, more precisely, to a specific device. This paper asks the only question that matters: **under what conditions does a device stop needing a battery at all and invest harvest power ambiently?**

The answer does not come from a single breakthrough. It comes from the intersection of independent trends, each progressing on its own timeline — and, critically, one of them is already moving on its own, for reasons that have nothing to do with wireless power.

2. Design Philosophy

One vision of wireless power imagines kilometers-wide rectenna arrays and beams actively tracked onto individual devices as they move through space. This paper rejects that vision outright.

A system that requires dedicated infrastructure to track and aim energy at every device does not scale the way information networks scale. It recreates, for power, the exact point-to-point fragility that packet-switched networks were built to eliminate. It is also an invasive, utilitarian — A future in which billions of devices require continuously tracked energy beams is not only operationally complex—it is, frankly, a cringe-inducing vision of what abundant infrastructure should look like. Point-to-point power delivery is the architecture of scarcity. Ambient infrastructure is the architecture of abundance. The future of wireless power depends less on transmitting more energy than on choosing the right architecture.

This paper's governing principle:

Energy is ambient infrastructure, not an actively managed resource.

Information networks already prove this model works. A phone does not negotiate a dedicated optical path to load a webpage — it exists inside a pervasive field of coverage and pulls what it needs, imperfectly, without tracking. Power follows the same model. The mechanism that makes it possible is the one beam-focused architectures spend their engineering budget eliminating: **diffraction**.

Where a tracked beam minimizes diffraction to keep energy collimated onto a single receiver, this architecture uses diffraction to fill a space with usable, low-intensity energy — energy that many devices, at arbitrary and unpredictable positions, harvest simultaneously without any of them being individually targeted.

This has a direct brutal tradeoff, and it is the cost that defines this entire paper: **power density at any single device drops dramatically**. There is no way to engineer around this. It follows from the physics of energy spreading once it is no longer being steered.

This is why the paper's core argument exists. If ambient power density is inherently low, ambient-powered computing is not reached by improving transmission or harvesting efficiency alone. The equation only closes if **device power requirements fall by orders of magnitude**. That is not a caveat. It is the thesis.

A traditional engineer will look at this and call it wasteful — most of the energy radiated into a space is never captured. **This objection uses the wrong metric**. Efficiency matters when energy is scarce and the goal is moving a large, specific quantity of power from one point to another. Ambient-powered computing is not moving a kilowatt anywhere. Once device power consumption has fallen far enough, the only question that matters is whether there was enough — and the answer is yes, even when most ambient energy in a space goes uncaptured. We do not call Wi-Fi inefficient because a router radiates in every direction and only a fraction reaches any device. We do not call sunlight wasteful because most of it lands on pavement. Ambient infrastructure is generous by nature, not precise — and that stops being a problem the moment the receiving side stops needing very much.

The future this paper describes is one where power, like signal, simply fills a space — abundant enough in aggregate, low-intensity enough to be safe, and modest enough per device that capturing it requires no tracking, no aiming, no optimization toward a specific point. That threshold is microwatts for one device and milliwatts to a few watts for another. The number is not the point. **The point is that it is bounded, known, and small enough to fall within what a diffuse ambient field provides**. The question of waste only makes sense when the target is still measured in kilowatts. Ours is not.

3. Physics Does Not Disappear

This paper's conviction is not a suspension of physics. Every claim here operates within constraints that are decades old and non-negotiable.

The inverse-square law. Power density falls with the square of distance from any source. No engineering avoids this.

Diffraction. Electromagnetic energy spreads and bends rather than traveling as a contained beam. Beam-tracking treats this as a nuisance. This paper uses it as the mechanism that fills a volume of space with usable ambient energy. The physics is identical either way — only the design intent differs.

Aperture limits. A receiver's effective aperture bounds how much energy it can collect, as a function of antenna geometry and wavelength. Smaller devices have smaller apertures and collect less power. This is a hard constraint on miniaturization, full stop.

Friis transmission. The classical relationship between transmitted power, distance, wavelength, and gain gives a first-principles estimate of what arrives at a receiver in free space. Real ambient environments are diffuse and multipath-rich, so the idealized equation is a lower-bound sanity check, not a final model.

Rectenna efficiency. Conversion efficiency drops as incident power density drops — precisely the regime ambient harvesting operates in. This compounds the challenge. It does not eliminate it.

Power density. Ambient RF power density in ordinary environments is typically many orders of magnitude below what a focused beam achieves. This is the central fact. This paper does not argue it away.

These constraints are not obstacles to route around. They are the reason this paper's argument takes the shape it does. They rule out any version of ambient-powered computing that requires diffuse fields to behave differently than physics allows. They do not rule out the one path that remains compatible with them: designing devices whose power requirements fall to meet the field, rather than fields engineered to meet the device.

4. The Fundamental Inequality

Most wireless power research asks: **how much power can be transmitted?**

This is the wrong question for ambient-powered computing. It treats only one side of the problem as variable, and bottlenecks progress entirely on transmitter design and regulatory limits — both slow to move, some immovable.

The right question: **when does harvested ambient power exceed what a device actually requires?**

This converts a single-variable optimization into a two-sided inequality:

Harvested Power $\geq$ Average Device Power Consumption

Harvested Power is a function of ambient power density, effective collecting area, and rectenna efficiency at that density. **Required Power** is a function of display, semiconductor efficiency, radios, sensing, and local computation.

For decades, research effort has gone almost entirely into one side of this inequality — larger transmitters, tighter beams, higher regulatory ceilings. That is a mistake. Section 3 already showed why: every constraint on the harvesting side is a hard physical limit or a slow materials science problem. The other side — Required Power — is a problem this industry has already been solving, incrementally, for fifty years, and continues to solve for reasons that have nothing to do with wireless power at all.

This is the core argument: the goal is not to “beam more power”. It is to close the gap from whichever side is moving faster — and one side is already moving on its own and will continue to do so.

That is not a hope. It is the pivot to Section 5, and it is why the timeline for ambient-powered computing is set not by breakthroughs in RF harvesting, but by a curve that is already in motion for entirely unrelated reasons.

5. Four Converging Curves

Section 4 established that ambient-powered computing depends on closing the gap between two sides of an inequality. This section names the four forces that close it — and argues that the fourth curve is the one that turns this from a terrestrial curiosity into a planetary-scale infrastructure bet.

Curve 1: Device Power Demand ↓. Already falling, driven by efficient chip architectures, lower-power displays, microwatt-class sensors, and aggressive power management. This curve requires no new physics — it is the electronics industry doing what it has always done, for markets that have nothing to do with ambient power.

Curve 2: Rectenna Efficiency ↑. Bounded by the physics in Section 3 — materials science, metasurface design, fabrication, impedance matching at low incident power density. The one curve of the four that depends on genuine research progress rather than market momentum alone.

Curve 3: Compute Decentralization ↑. Compute is leaving the device. Data centers and AI infrastructure — built by the largest companies on earth, at a capital scale with

no precedent outside a handful of historical infrastructure cycles — are absorbing the computational load that used to sit locally on-device. This is not a hypothetical trend. It is hundreds of billions of dollars of present-day capital expenditure, and every dollar of it pushes Curve 1 down further, for reasons that have nothing to do with ambient power. Section 7 makes the case that this is not incidental — it is the same capital pattern that has repeated every time infrastructure preceded its own killer application.

Curve 4: Launch Costs ↓. Reusable rocket architecture has already cut the cost of reaching orbit by an order of magnitude, and the trajectory continues. This curve is the one most people evaluating ambient-powered computing ignore entirely, because they are still thinking about wireless power as a terrestrial problem. It is not. **Falling launch costs are what convert orbital power infrastructure from a physics thought experiment into a capital-efficient deployment.** A satellite constellation beaming diffuse, low-intensity power toward the earth's surface is not science fiction gated on some undiscovered propulsion breakthrough. It is gated on cost per kilogram to orbit — and that cost is already collapsing, driven by a private space industry that, like Curve 3, is not spending this capital because of ambient power at all.

The hypothesis: these four curves intersect, and ambient orbital power stops being speculative the moment they do.

This is not a coincidence of four unrelated trends. It is the direct, connectable consequence of where global capital is already flowing. Curve 1 is the semiconductor industry's default trajectory. Curve 3 is the AI infrastructure buildout every major tech company is currently executing in public. Curve 4 is the reusable-rocket economics that a handful of well-funded companies have already proven out. Curve 2 is the only piece that requires a dedicated research push — and it is the smallest, most tractable gap of the four.

6. Curve 3: The Decentralization of Compute

Reducing device power does not mean reducing device capability. It means compute leaves the device entirely — and this is not a projection, it is the current, visible strategy of the largest capital allocators in technology.

Devices become sensors, displays, microphones, cameras, and radios. The computationally expensive work happens in centralized data centers and increasingly in localized micro-data-centers positioned closer to where inference is actually needed — neither of which carries the device's power constraint. A device built this way consumes a fraction of the power a general-purpose computing device requires, without sacrificing

a fraction of the perceived capability, because perceived capability was never bottlenecked by local compute to begin with.

This is already happening, at a scale that makes it undeniable. The current AI infrastructure buildout — hyperscale data centers, dedicated inference clusters, purpose-built accelerator silicon — is being financed by the handful of companies with the largest balance sheets on earth, and it is being financed specifically because compute is migrating away from the edge and toward centralized infrastructure. Every dollar of that buildout is a dollar spent making Curve 1 fall faster, for reasons that have nothing to do with ambient-powered computing. This paper does not need that capital to be redirected. It only needs it to keep flowing in the direction it is already flowing.

Ultra-efficient local processors will preserve more on-device intelligence than a pure-decentralization model assumes — that is not a threat to this argument, it is a second lever pointed in the same direction. What ambient-powered computing requires is narrower: the device's irreducible local functions — sensing, minimal signal processing, radio — must fall within what the ambient field supplies. Whether that reduction comes from offloading to the data centers described in Section 7, or from local silicon becoming radically more efficient, is irrelevant to the inequality in Section 4. Both push the same variable in the same direction.

7. Capital Cycles

Capital consistently overbuilds infrastructure before the application that justifies it is known. This is not the exception. It is the pattern — and it is the pattern this paper's entire fourth curve depends on.

Railroads were laid into regions with no existing demand, financed on the bet that demand would follow capacity. It did, through industries the original capital never anticipated. Electrification followed the same arc: grids built for lighting and factory motors became the substrate for an entire consumer economy the initial business case never included. Fiber-optic buildouts in the late 1990s overbuilt capacity so aggressively that most of it sat unused for years — and that excess became the exact substrate that made streaming, cloud computing, and mobile broadband viable a decade later.

The AI infrastructure buildout: Hundreds of billions of dollars are currently being committed by the largest technology companies on earth to data centers, accelerator silicon, and inference capacity, at a scale that no currently known application fully justifies on its own. This is not speculation — it is quarterly capital expenditure, disclosed and public. It is the railroad and fiber pattern, replaying in real time, and it is

the direct engine behind Curve 3: every dollar of that buildout decentralizes compute away from the device and pushes device power requirements down.

History says the infrastructure arrives first and the highest-value application is discovered afterward. Data centers are already being built. The application this paper is describing is one of the things they make possible.

8. Research Agenda

Device power (Curve 1): How low can power realistically fall before further reduction costs capability rather than efficiency? Is there a practical floor set by radio power alone, once compute and display fall further?

Harvesting efficiency (Curve 2): How high can rectenna efficiency reach specifically in the low-power-density regime ambient environments — terrestrial and orbital-beamed — operate in? Do metamaterial approaches meaningfully increase effective aperture without increasing physical size? What is the realistic manufacturing cost at consumer volume?

Compute decentralization (Curve 3): How much further can data-center and edge-inference buildout push local device power requirements down? Which local computation resists offloading, and does that set shrink as localized micro-data-centers proliferate?

Launch costs (Curve 4): At what cost-per-kilogram-to-orbit does a diffuse orbital power constellation become capital-competitive with terrestrial ambient infrastructure? What orbital architecture — altitude, constellation density, beam diffusion pattern — maximizes usable ground-level power density while remaining within safety and regulatory exposure limits? How does the reusable-launch capital cycle's trajectory compare to the AI infrastructure buildout's trajectory, and which one is the binding constraint?

Cross-cutting: Can battery-free mobile robotics exist, given that actuation carries power requirements sensing and compute do not — and does this require energy buffering rather than pure real-time harvesting? What regulatory and safety frameworks govern power density beamed from orbit toward populated areas, and how do they compare to existing terrestrial RF exposure limits? How should cities integrate receiving infrastructure for orbital ambient power from first principles, rather than retrofit it?

9. Conclusion

Ambient-powered computing is not a wireless power transmission problem. It is a four-curve convergence problem — and three of the four curves are already moving without anyone building them for this reason.

Device power is falling because the semiconductor industry has been falling for fifty years. Compute is decentralizing because the largest technology companies on earth are currently spending hundreds of billions of dollars to make it so. Launch costs are collapsing because reusable rocket economics have already been proven out by companies with no interest in ambient power whatsoever. Only rectenna efficiency — Curve 2 — remains a dedicated research problem, and it is the smallest gap of the four.

This is the argument in full: connect a semiconductor roadmap, a hyperscaler capex chart, and a launch-cost curve, and ambient orbital power is not a bet on a future breakthrough. It is the visible intersection of three capital cycles already underway, plus one tractable research gap.